Homework — Stage 09 Feature Engineering

All paths below are inside `homework/homework09/`.

Assignment

In the lecture, we learned how to create new features from raw variables.
Now, you will adapt those ideas to your own project data.

Tasks

1. Create **at least 3** engineered features in your dataset. At least one must encode a categorical column - one-hot, label, or frequency (the lecture demonstrates all three on `region`).
    - Each feature must include:
        - Implementation code
        - Markdown explanation of reasoning
        - Visualization or correlation check
2. Save your work as **both**:
    - `homework09_feature-engineering_submission.ipynb`
    - AND your helper module `src/features.py`
3. Commit and push to your repo before the next session.

Step-by-Step Instructions

1. Copy `stage09_feature-engineering_homework-starter.ipynb` into `homework/homework09/`, rename it `homework09_feature-engineering_submission.ipynb`, and open it.
2. Replace the sample synthetic data with your project dataset.
3. Implement at least 3 new features, one of them a categorical encoding.
4. Document each with a short rationale, naming the stage 08 EDA insight it comes from.
5. Plot or test the correlation between each feature and the target variable.
6. Save and commit.

Grading Rubric (100 points)

- **20 points**: At least three features, one of them a categorical encoding
- **20 points**: Each feature documented with clear rationale
- **20 points**: Features connect to EDA insights
- **20 points**: Code is correct and reproducible
- **20 points**: Plot or correlation check, with a sentence on what it tells you

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Example Submission Expectation

- Feature 1: `spend_income_ratio`
    - Code snippet: `df['spend_income_ratio'] = df['monthly_spend'] / df['income']`
    - Rationale: “Helps capture proportionality of spending to earning.”
- Feature 2: `rolling_spend_mean`
    - Code snippet: `df['rolling_spend_mean'] = df['monthly_spend'].rolling(3).mean()`
    - Rationale: “Captures 3-month trend in spending behavior.”